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Mechanisms of brain activation following naming therapy in aphasia: a systematic review on task-based fMRI studies

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ABSTRACT

The pattern of brain neuroplasticity after naming therapies in patients with aphasia can be evaluated using task-based fMRI. This article aims to review studies investigating brain reorganization after semantic and phonological-based anomia therapy that used picture-naming fMRI tasks. We searched for those articles that compared the activation of brain areas before and after aphasia therapies in the *PubMed* and the *EMBASE* databases from 1993 up to April 2020. All studies (single-cases or group designs) on anomia treatment in individuals with acquired aphasia were reviewed. Data were synthesized descriptively through tables to allow the facilitated comparison of the studies. A total of 14 studies were selected and reviewed. The results of the reviewed studies demonstrated that the naming improvement is associated with changes in the activation of cortical and subcortical brain areas. This review highlights the need for a more systematic investigation of the association between decreased and increased activation of brain areas related to anomia therapy. Also, more detailed information about factors influencing brain reorganization is required to elucidate the neural mechanisms of anomia therapy. Overall, regarding the theoretical and clinical aspects, the number of studies that used intensive protocol is growing, and based on the positive potential of these treatments, they could be suitable for the rehabilitation of people with aphasia.

KEYWORDS

Anomia therapy; brain reorganization; neuroplasticity; phonological-based treatments; picture naming fMRI task; semantic-based treatments

Introduction

Aphasia is an acquired communication disorder caused by damage to the brain areas responsible for formulating and comprehending language and its components (Mack et al., 2013). Approximately 60% of persons with aphasia (PWAs) still experience language symptoms in the chronic phase of the disorder starting 6 months post-onset (Meinzer et al., 2005). The most common chronic symptom of aphasia is the inability to retrieve specific words or anomia that is present in all types of the disorder and at various severity levels (Azhar et al., 2017; Flöel et al., 2011; Maher & Raymer, 2004; Nardo et al., 2017). Studies focused on anomia for improving naming abilities, language and communicative function, and even quality of everyday life of PWAs (Best et al., 2008; Boyle, 2004; Boyle & Coelho, 1995; Coelho et al., 2000; Davis & Stanton, 2005; Doogan et al., 2018; Efstratiadou et al., 2018; Kristensson et al., 2015; Mehta & Isaki, 2016; Peach & Reuter, 2010; Silkes et al., 2019; Simic et al., 2021; Wallace & Kimelman, 2013).

According to the lexical models of word production, anomia arises from impairment in the semantic or phonological

level of word processing or at the connection between these two levels (Dell et al., 2007; Foygel & Dell, 2000; Schwartz, 2014, 2006). Semantic-based and phonological-based treatments of anomia are two standard methods of anomia therapy that focus on semantic and phonological aspects of word retrieval processing, respectively (Abel et al., 2005; Best et al., 2002; Boyle & Coelho, 1995; Conroy et al., 2009; Hickin et al., 2002; Kiran et al., 2013; Leonard et al., 2008; Lorenz & Ziegler, 2009; Lowell et al., 1995; Madden et al., 2017; Nettleton & Lesser, 1991). However, studies have shown that semantic-based treatments affect word form activation and phonological-based treatments influence semantic processing; therefore, both treatments activate all levels of word retrieval processing (Leonard et al., 2015, 2008; Marcotte et al., 2018; Neumann, 2018; van Hees et al., 2013).

A growing number of studies have assessed the roles of the above-mentioned therapy methods in treatment-induced neural plasticity (Davis & Harrington, 2006; Dressel et al., 2010; Heath et al., 2012; Leonard et al., 2015; Meinzer et al., 2007; Perani et al., 2003; Rochon et al., 2010). Knowing how

the brain responds to this modification will help to improve our understanding of the brain's compensatory systems to define more accurate treatments and rehabilitation techniques (Nickels et al., 2011; 2015; Small et al., 2013). Recognition of these mechanisms also is an essential method for investigating the neural basis of cognition (Nickels et al., 2011, 2015). For this purpose, functional and structural neuroimaging are the most suitable methods (Crosson et al., 2019; Lee et al., 2006).

As a functional imaging technique, Task-based functional magnetic resonance imaging (task-based fMRI) studies have been developed to identify brain states during various processing conditions (Matthews & Hampshire, 2016). Task-based fMRI could be used to measure activation in specific brain areas (rather than multiple) and find out patterns of neural network reorganization in a specific domain such as language (Carter et al., 2012; Meinzer et al., 2011; Smits et al., 2012; van Meer et al., 2012; Zhang et al., 2013). As rehabilitation practices are associated with changes in brain activation and connectivity, therefore growing studies have used this method for investigating mechanisms of rehabilitative treatments (Crosson et al., 2019; Dobkin, 2004; Kiran & Thompson, 2019; Wilson & Schneck, 2021). In task-fMRI used for treatment-induced neuroplasticity, changes in neural activation may indicate increased neural processing (i.e., more naming effort representing more BOLD signal) or may be associated with behavioral enhancements (Kiran et al., 2013). Therefore, Knowing the cognitive aim of a task, understanding the relationship between fMRI task and neuronal activation, the sensitivity of the selected task to neural mechanisms of rehabilitation (Crosson et al., 2007; Heeger & Ress, 2002; Kiran et al., 2013; Saur & Hartwigsen, 2012), and also knowing the theory of therapy are prerequisites for investigation the mechanism of treatment-induced neuroplasticity (Meinzer & Breitenstein, 2008; Nickels et al., 2011, 2015).

Most studies of the neural basis of language production use picture naming tasks for studying word retrieval processing (Dell et al., 2007; Laine & Martin, 1996; Meinzer & Breitenstein, 2008). Visual stimuli are first converted to a conceptual representation for naming a picture. Then, semantic, lexical, and phonological representations provoke picture name retrieval, and finally, phonetic encoding and motor activation contribute to the articulation of that name (Cuetos et al., 2000; Dell et al., 1997). For instance, to name a picture of a specific horse as "horse," one requires to at least: (1) recognize the visual stimulus as an example of a familiar concept; (2) have access to the meaning of the horse, (3) have access to the pronunciation of the word, and (4) prepare speech motor plan for speech articulation (DeLeon et al., 2007). Because semantic and phonological-based treatments focus on picture naming processing levels, using the picture naming task in fMRI is appropriate for investigating the neural mechanism of word retrieval treatments. Therefore, this study aimed to review the functional changes in brain areas related to semantic and phonological-based treatments, which were assessed by picture naming tasks.

Methods

We performed this systematic review by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Liberati et al., 2009). The registration number of this review in the PROSPERO database (international registry of progressively registered systematic reviews) is CRD42020198140.

Data sources

PubMed and EMBASE databases were systematically searched from 1993 up to April 2020. The search was conducted with the following keywords with different combinations of (And) and (Or):(((((((therapy [MeSH Terms]) OR therapy[Title/Abstract]) OR treatment[Title/Abstract])) AND ((aphasia[MeSH Terms]) OR aphasia[Title/Abstract])) AND (((stroke[Title/Abstract]) OR CVA[Title/Abstract]) OR "Cerebrovascular Accident"[Title/Abstract]) OR stroke[MeSH Terms])) AND (((fmri[Title/Abstract]) OR "functional MRI"[Title/Abstract]) OR "functional magnetic resonance imaging"[Title/Abstract])). Also, a manual search for the references of the selected studies was carried out to cover probably missing articles. This additional search was necessary, as not all studies are indexed in databases.

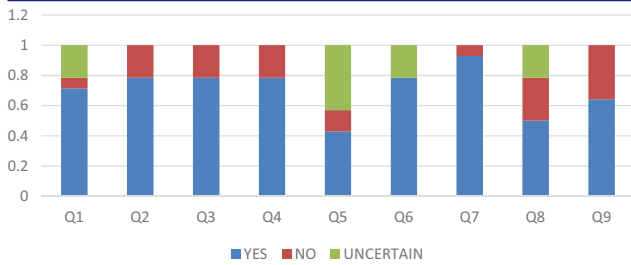
Study selection

Selections were performed by two independent reviewers screening titles and abstracts. In cases of ambiguity within the abstract, full texts were considered. Disagreements were argued with a third reviewer to make the final decision. The inclusion criteria were as follows: (1) Functional MRI studies containing any type of aphasia caused by CVA. (2) Studies that used any treatments to alleviate the aphasia. The studies with the following characteristics were excluded: (1) Rats and other animal studies, (2) Aphasia caused by progressive neurological diseases, (3) Patients with other neural disorders/diseases. (4) Studies that focused only on functional connectivity. (5) Studies that did not use semantic and phonological-based treatment. (6) Studies that did not use pre-post picture naming fMRI task. (7) Studies used tDCS, TMS, or rTMS as the treatment protocol. (8) Studies published in any language other than English. (9) Studies without available full text.

Data extraction

From the selected studies, the study design, participants' demographic information and clinical characteristics (including mean age, gender, type of aphasia, handedness, education, lesion characteristics, and post-stroke time), MR Imaging acquisition and analysis characteristics (including magnetic field strength, software tools and packages, brain atlases and fMRI tasks), aphasia assessment test characteristics (assessment test types, scores before and after treatment), treatment characteristics (the type of treatment protocol and the number of sessions) and alterations in

Table 1. JBI critical appraisal checklist for quasi-experimental studies (i.e., non-randomized experimental studies).



brain activation post-treatment were extracted. Data were synthesized using a narrative method and meta-analysis was not conducted due to the heterogeneity of the studies (Table 1).

Quality appraisal

The quality assessment was performed by utilizing the Joanna Briggs Institute (JBI) critical appraisal tool (<https://jbi.global/critical-appraisal-tools>) for quasi-experimental studies. Two reviewers (F. D. and A. M.) independently assessed the risk of bias in each study. They discussed disagreements on selected articles and the quality assessment results, and decisions were made based on discussion and consensus. Quasi-experimental studies were evaluated based on nine criteria. All questions were answered as yes, no, not clear, or not applicable and assessed individually. Each Eligible study was rated according to the dictionary and guidelines of the tool. After evaluating all the components of the study, the overall rating was determined using the criteria defined in the tool. Based on the number of “yes” responses, a rating of good = (≥ 7 yes), medium = (4–6), and poor = (≤ 3 yes) was assigned for the nine questions.

Results

Included studies

A total of 14 studies were included (Figure 1). 146 aphasia participants (63 females) participated in these studies. 11 studies had reported detailed data about the age of participants that ranged from 21 to 81 years (Mean = 56.04 ± 11.91). The reported post-stroke time ranged from 5 to 350 months (Mean = 59.24 ± 64.71). The demographic data of each article’s participants are presented in Table 2. All participants of the reviewed studies were in the chronic phase of stroke. Although some studies did not report handedness, all participants were right-handed for the reminded studies. However, the reviewed studies did not report random sampling, blindness, or treatment fidelity. None of the studies was conducted on multilinguals.

Participants have different types and severities of aphasia and diverse word-finding abilities (Tables 2 and 3). The most frequent types were Broca’s and anomic aphasias. The severity of participants’ aphasia ranged from mild to severe. All studies have reported that their patients have significant

word-retrieval problems or they focused on naming problems. We tried to report the intensity of treatment, i.e., the number of implemented sessions during a specific period. The average duration of the treatment period was 4.4 (SD 2.9) weeks. The detailed information about the language assessments, therapy techniques, and treatment program is presented in Table 3.

Most studies (11) used 3 T MRI scanners, and the rest employed 4 T or they did not describe the scanner characteristics. The methods of studies have been summarized in Table 3. All studies used T1-weighted images with appropriate voxel size, although their utilized packages for the volumetric analysis were different. Nine studies used pre- and post-test assessment group designs, and five used single-subject designs. The fMRI task of all reviewed studies was picture naming.

Behavioral naming improvement

Seven studies alternatively used semantic and phonemic-based protocols, 4 studies used phonological-based treatment, and 3 studies used semantic-based treatment. All studies showed significant naming improvement for treated items. The generalization effects for untreated items were also observed in six studies. 98 participants showed naming improvement for treated items, and 63 participants showed naming improvement for untreated items. However, the treatment effects were negligible for untreated items.

Relation between naming improvement and change in brain activation

The reviewed studies showed that naming improvement is associated with changes in activation of brain areas. Although most studies (11) reported that naming improvement is associated with increased activation of brain areas, five studies also showed that decreased activation could be associated with naming improvement. As provided in Tables 4 and 5, broad areas in both left and right hemispheres are associated with naming improvement. Also, the total number that each area is reported in the reviewed studies is provided in Figures 2 and 3. These data could estimate the degree of plasticity of the region following aphasia recovery. We also provided the maps of brain areas related to naming improvement (Figures 4–7).

Discussion

This evidence-based systematic review aimed to examine the roles of brain areas in neuroplasticity induced by anomia therapy. Semantic-based and phonological-based treatments are two common methods of anomia therapy that focus on semantic and phonological processing, respectively. However, it seems that semantic treatments affect phonological processing abilities and vice versa (Leonard et al., 2015, 2008; Marcotte et al., 2018; Neumann, 2018; van Hees et al., 2013). Therefore, we analyzed fMRI studies that used

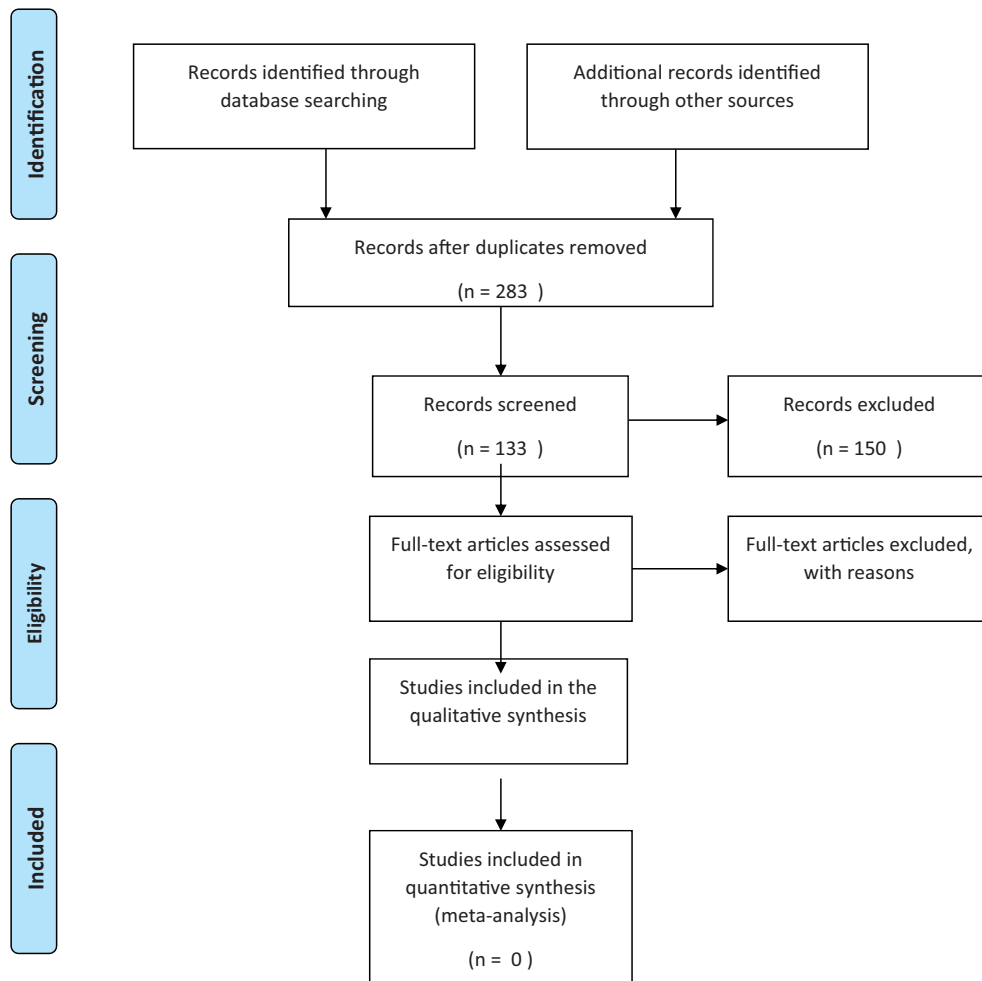


Figure 1. Prisma flow diagram.

Table 2. Demographic information of participants.

Study	Participants	Age (years)	Gender	Handedness	Education	Months post onset
(Kiran et al., 2015)	8	Mean = 58	1 F	Not mentioned	Not mentioned	85 (Range 15 – 157)
(Vitali et al., 2007)	2	24/53	0 F	Right	8/13	12/ 48
(Abel et al., 2014)	14	Median = 48	4 F	Right	10.8	41
(Abel et al., 2015)	14	Median = 48 (range = 35 – 74 years)	4 F	Right	10.8	Median time post-onset = 41 months (11 – 72)
(Menke et al., 2009)	8	Range = 34 – 67	3 F	Right	Not accessible (file was not found)	Not accessible (file was not found)
(Fridriksson, 2010)	26	Mean = 59.7 (SD ± 12.3)	14 F	Not mentioned	Not mentioned	8 months
(Fridriksson et al., 2012)	30	Range = 33 – 81 years; mean = 59.2 years	16 F	Not mentioned	Not mentioned	51.1 months, range = 6–350 months
(Fridriksson et al., 2006)	3	62/68/47	3 F	Not mentioned	MSc/high school	84/144/24
(Fridriksson et al., 2007)	3	63, 42, 63	2 F	Not mentioned	Not mentioned	12/ 22/ 98
(Nardo et al., 2017)	18	Mean = 50 (SD ± 12)	6 F	Right	Not mentioned	Mean = 61 (SD ± 58)
(Marcotte & Ansaldo, 2010)	1	66	1 M	Right	18	96
(Marcotte et al., 2012)	9	Mean = 62 (SD ± 6)	4 F	Right	14.8	Mean = 110.2 SD ± 92.5
(Marcotte et al., 2018)	2	59 58	1 F	Right	Not mentioned	36 12
(van Hees et al., 2014)	8	Mean = 56.38; SD ± 9.15 Range = 41 to 69	5 F	Right	Not mentioned	Mean = 52.25; SD ± 49.8 Range = 17 to 170

SD: standard deviation.

Table 3. Study designs and language performances of the participants and overall quality assessment.

Study	Study design	Language assessment	Aphasia type	Word— finding severity		Treatment (s)	Session	Naming improvement	Overall Quality
				Mild to severe	finding severe				
(Kiran et al., 2015)	Pre- and post-treatment group analysis	WAB, PPT, CLQT	Not mentioned			SFA	Weekly until reaching 80 percent accuracy or higher	All patients improved after rehabilitation for treated categories (medium to large effect sizes). Improvement on untreated items also was observed.	Good
(Vitali et al., 2007)	Single-subject	Snodgrass and Vanderwart picture set	2 Global	One severe / one moderate		Phonological cues	8 weeks for p 1; 4 weeks for p 2	Both patients showed improved naming of the treated list, without generalization to the untreated items	Poor
(Abel et al., 2014)	Pre- and post-treatment group analysis	AAT	1 Transcortical sensory 3 Wernicke's 1 fluent non-classifiable 1 Global 8 Broca's	Moderate—severe		Model-oriented / semantic / phonological therapy	4-week	Significant improvement for treated and untreated items in 9 participants. (Large effect size for treated items and medium effect size for untreated items.)	Good
(Abel et al., 2015)	Pre- and post-treatment group analysis	AAT				semantic or a phonological cueing	4-week	Significant naming improvement	Good
(Menke et al., 2009)	Pre- and post-treatment group analysis		7 Broca 1 global	Moderate to severe		Phonological cues, Grapheme cues, semantic cues	Daily for 3 h for 2 weeks (intensive)	Significant improvement for treated (in 7 participants) and untreated items (in 6 participants) (improvement was more for treated items)	Good
(Fridriksson, 2010)	Pre- and post-test group analysis	WAB	11 Anomic, 10 Broca's, 3 conduction, one Wernicke's, one TCM	Not mentioned		A semantic- or phonologic-based cueing	2 weeks of intensive language treatment (With three exceptions, each patient received 3 h of per weekday for 2 weeks for a total of 30 h) 3 patients (2 hours per day) Three hours of anomia treatment per weekday for 2 weeks, for a total of 30 h	Naming improvement for 18 patients Significant improvement at the group level.	Good
(Fridriksson et al., 2012)	Pre- and post-test group analysis	WAB	13 Broca's, 10 anomic, 3 conduction, 2 Wernicke's, 1 TCM, and 1 global	Not mentioned		Semantic- or phonologic-based cueing		Naming improvement for 16 aphasia patients.	Good
(Fridriksson et al., 2006)	Multiple single-subject design	BDAE /BNT	Broca's / Anomic	2 Severe/ one Mild		Phonemic cues, repetition, naming	4 h per day for 2 weeks	Significant improvement for	Good

(continued)

Table 3. Continued.

Study	Study design	Language assessment	Aphasia type	Word— finding severity	Treatment (s)	Session	Naming improvement	Overall Quality
(Fridriksson et al., 2007)	Single-subject design	WAB/PNT	3 (2 Non—fluent, one fluent) Conduction, Broca	Not mentioned	semantic or phonological treatments	a total of ten 4-h treatment sessions) five sessions phonemic cueing five sessions semantic cueing. Each session lasted 2 h, and each approach was employed for five consecutive days 2 h of naming practice daily over 6 weeks Mean 73 h (25)	two patients and poor improvement for one patient Naming improvement and generalization for two patients and decreased anomia for one patient.	Medium
(Nardo et al., 2017)	Pre- and post-test group analysis	BNT/ CAT/PALPA	Not mentioned	Not mentioned	Phonemic cues		Significant naming improvement for treated (17 patients) and untreated items (13 patients)	Good
(Marcotte & Ansaldi, 2010)	Single-subject design	M-Tb	Broca aphasia	severe aphasia	SFA	Three 60-min sessions per week for 3 weeks	Significant improvement for treated and untreated items	Medium
(Marcotte et al., 2012)	Pre- and post-test group analysis	M-Tb	8 Broca One Wernicke	1 moderate 4 moderate-to-severe 4 severe	SFA	For an hour and was provided three times a week (a mean of 4.3 weeks and an average of 12.774,2 therapy sessions)	Significant naming improvement for treated and untreated items in all patients.	Good
(Marcotte et al., 2018)	Pre- and post-test analysis in a single- subject design	BNT/BORB/WAB	2 Broca	2 moderate-to-severe	PCA	Intensive treatment for P2, (10 sessions of 3 h each, in of 2.5 weeks, P1 standard treatment, (30 sessions of 1 h each, in 10 weeks)	Improvement for treated items but not for untreated items. Significantly (only for one patient)	Poor
(van Hees et al., 2014)	Pre- and post- treatment group analysis	WAB /BNT/PPT/CAT	2 conduction 6 anomic	5 mild 1 mild-to-moderate 2 moderate	SFA PCA	12 therapy sessions (three sessions per week for 4 weeks)	Naming improvement for 7 patients on treated and untreated items	Good

Language assessment test: AAT: The Aachen Aphasia Test; BDAE : Boston diagnostic aphasia; BNT: Boston naming test; BORB : Birmingham Object Recognition Battery; CAT : Comprehensive Aphasia Test; CLQT : Cognitive Linguistic Quick Test; M- Tb : Montreal-Toulouse 86 Beta version; PALPA : Psycholinguistic Assessments of Language Processing in Aphasia; PPT : Pyramids and Palm Trees; WAB : western aphasia battery. Treatment protocol: PCA : phonological component analysis; SFA :semantic feature analysis.

Table 4. Imaging information of the reviewed studies.

Study	Lesion type	Lesion location	Lesion size Mm ³	Decreased activation in left hemisphere related to naming improvement	Decreased activation in right hemisphere related to naming improvement	Increased activation in the left hemisphere related to naming improvement	Increased activation in the right hemisphere related to naming improvement	fMRI sessions and acquisition intervals
(Kiran et al., 2015)	7 left ischemic 1 right ischemic	Frontoparieto-temporal areas	214.56 mm ³			SFG, IFG, MFG, PoCG, SFG,	MFG, MTG, IFG,	2 sessions one in the baseline phase (pretreatment scan) and one in the post- rehabilitation phase (post-treatment scan)
(Vitali et al., 2007)	Closed Head injury/ Left Ischemic Stroke MCA	Frontoparieto-occipital/ fronto-temporo-parietal						2 sessions a pretraining er-fMRI session and a posttraining er-fMRI 2 sessions
(Abel et al., 2014)	13 ischemic	B.G /parietal, temporal/ frontal / insula		STG, precuneus, PCG, thalamus (pulvinar).	PoCG, Caudate, MCG, thalamus			
(Abel et al., 2015)	1 ischemic—hemorrhagic 13 ischemic 1 ischemic—hemorrhagic	B.G /parietal, temporal/ frontal / insula		IFG (opercularis), SMA, MCG, MFG, PCG, PoCG, IPL, precuneus, cerebellum	MTG, STS; thalamus, hippocampus, PCG, MCG, SMA PaCL, precuneus, cerebellum,	MFG, SFG, IFG (triangularis), Insula, SMG, MTG, SMG.	Precuneus, PCG, Hippocampus	
(Menke et al., 2009)	Left 7 Ischemic 1 hemorrhagic	B.G /parietal, temporal/ frontal / insula				Parahippocampi, hippocampus, fusiform gyrus	Para hippocampi, precuneus,PCG, fusiform gyrus.	3 sessions The first prior to the training (pre), the second directly after the training (post1), and the third at an 8-month follow- up (post2).
(Fridriksson, 2010)	Ischemic	BA 37, BA 39, and BA 19, Insula, Broca, MTG				MFG, IFG (Parsopercularis), SPL, IPL, PCG, Precuneus,		6 sessions (Two sessions before treatment, two sessions during the 1-week treatment break, and two final sessions after the treatment).
(Fridriksson et al., 2012)	Ischemic and/or hemorrhagic	Frontoparieto temporal areas.	Mean = 115.91 (range = 7.7 –420.5)			IFG, SFG, MFG, (Perilesional areas)		4 sessions Two sessions before treatment and two sessions immediately after treatment 6 sessions for two patients and 5 sessions for one patient.
(Fridriksson et al., 2006)	Thromboembolic	Frontoparieto temporal areas.				Frontal pole, ACC, SMG, Temporal Pole, IPL		(Three baseline fMRI sessions and three post-treatment fMRI sessions)
(Fridriksson et al., 2007)	Ischemic	The IFG, insula, STG (BA 41/ 42), IPL (BA 39/40), (BA19), the basal ganglia (caudate, pallidum, and putamen)				SPL (Precuneus), SPL (Precuneus), entorhinal cortex, thalamus.		6 sessions (2 sessions before the first treatment, 2 sessions after completion of the first treatment phase, and two sessions after the end of the second treatment phase).

(continued)

Table 4. Continued.

Study	Lesion type	Lesion location	Lesion size mm ³	Decreased activation in left hemisphere related to naming improvement	Decreased activation in right hemisphere related to naming improvement	Increased activation in the left hemisphere related to naming improvement	Increased activation in the right hemisphere related to naming improvement	fMRI sessions and acquisition intervals
(Nardo et al., 2017)		Frontoparieto temporal areas.	161 (84)	PCG, SMA	Insula, ACG, IFG			2 sessions Before and immediately after therapy
(Marcotte & Ansaldo, 2010)	Ischemic	Left frontotem- poral				SFG, IOG, globus pallidus	PCG, SFG, IOG,	2 sessions (Before and immediately after therapy)
(Marcotte et al., 2012)		Frontal, parietal, temporal, insula	138.98 (range 84.77 – 295.76)			PCG, IPL, MTG, IFG, SMG, SFG.	PCG MTG, cuneus precuneus, STG, SFG,	2 sessions First fMRI before treatment, A second fMRI session at the end of the treatment period
(Marcotte et al., 2018)	Thromboembolic			SFG, MTG, PCG, Putamen,	PCG, PCG, PoCG, precuneus	SFG, STG, IFG,	Caudate, precuneus, STG, PCG	2 sessions (A baseline fMRI session ("pre" time point), and a second fMRI session ("post" time point)
(van Hees et al., 2014)	Ischemic	MTG, STG, IFG, SFG, Insula, SMG, putamen, culate, amygdala, thalamus, PCG, PoCG, SMA, ACC,	67.71	IFG (triangularis, opercularis), MTG, STG, IFG (orbitalis)	STG, MTG, IFG (orbitalis), IFG (Opercularis)	SMG, IFG (pars orbitalis, and triangularis), AG, ITG,	AG,	2 sessions one before training and second Following the therapy

ACC: anterior cingulate gyrus; AG: angular gyrus; FP: frontal pole; IOG: inferior occipital lobe; IPL: inferior parietal lobe; ITG: inferior temporal gyrus; MCG: middle cingulate gyrus; MOG: middle occipital gyrus; PaCL: para-central lobule; PoCG: post central gyrus; PCG: posterior cingulate gyrus; SMA: supplementary motor area; SMG: supramarginal gyrus; STS: superior temporal sulcus; TP: temporal pole.

picture-naming to identify each area's exact role in treatment-induced neuroplasticity related to these two treatments. By knowing the brain activation related to different anomia therapies and considering the potential factors related to this activation, we can choose optimal language therapy for each aphasia participant (Crinion et al., 2013; Crosson et al., 2007; Kristinsson et al., 2021).

The systematic search of literature since 1993, when the first fMRI studies on language functioning were reported (Hinke et al., 1993; McCarthy et al., 1993), was yielded 14 studies that provided data about the effects of semantic-based and phonological-based treatments on brain activation. Although one cannot interpret the results with certainty due to the extracted studies' limited number and experimental quality, this review revealed that both semantic and phonological treatments improved naming. Even in the chronic stage after a stroke, anomia treatment is associated with brain reorganization. Both left- and right-hemisphere areas are involved in naming improvement by increased and decreased activation of brain areas. Also, the treatment protocols, treatment intensity, the severity of word-finding difficulties, the impaired level of word retrieval processing, lesion location, and lesion size can affect the mechanism of neuroplasticity.

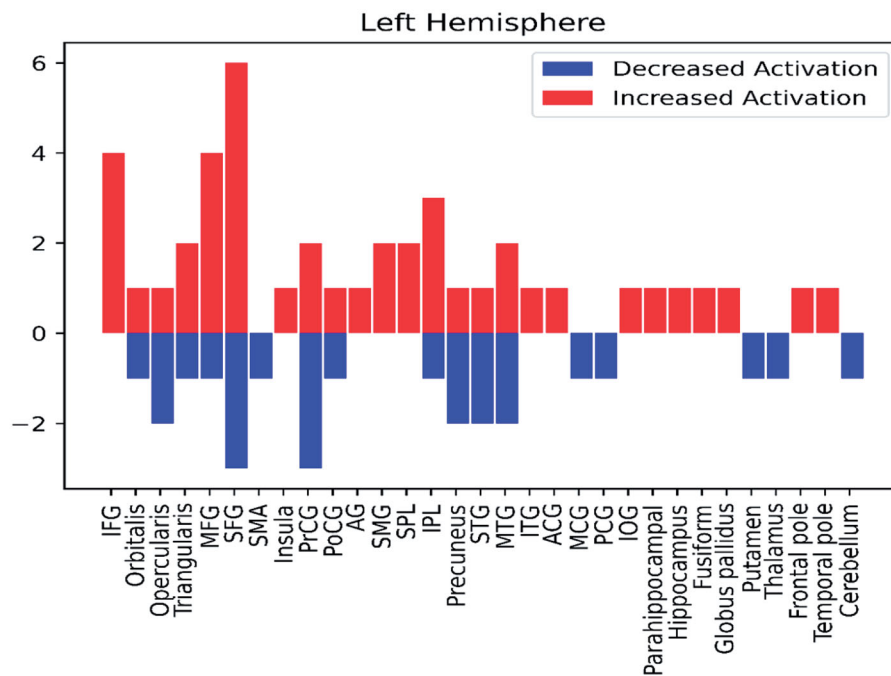
Relationship between the right and left hemisphere activation and naming improvements

Although both left and right areas are associated with naming improvement (Fridriksson et al., 2007; van Hees et al., 2014; Vitali et al., 2007), it has been shown that more significant improvement in successful naming is associated with a tremendous increase in left hemisphere cortical activation (Fridriksson et al., 2007, 2012; Marcotte et al., 2012; van Hees et al., 2014; Vitali et al., 2007). Several variables may also affect the activation of left, right, or both hemispheres regarding naming improvement. For example, Vitali et al. (2007) and Fridriksson et al. (2007) showed that **small lesion size is associated with more significant improvements after anomia therapy**. This improvement was related to the activation of left language-related areas. On the other hand, **larger lesion size was associated with slight naming improvement**. This improvement was related to the activation of the right areas. Also, some studies showed that naming improvement negatively correlated with damage to Broca's area (Abel et al., 2015; Marcotte et al., 2012; Vitali et al., 2007) and damage to posterior portions of the left temporal lobe (Fridriksson, 2010). Therefore, extensive damage to core language areas may restrict the naming improvement (Fridriksson et al., 2012; Menke et al., 2009). Providing more treatment sessions and hours in shorter periods, which is defined as extensity of therapy (Bhagal et al., 2003; Dignam et al., 2016; Sage et al., 2011; Stahl et al., 2018) may also explain the difference between the activation of the left or right hemispheres in naming improvement. For example, some studies showed significant improvement after intensive anomia therapy is associated with recruiting fewer brain areas in the left hemisphere after training (Fridriksson,

Table 5. Relationship between activated brain areas and treatment protocols.

Study	Treatment	Decreased left	Decreased right	Increased left	Increased right
(Vitali et al., 2007)	Phonological cues			MFG, SFG, IFG (triangularis), Insula, SMG, MTG, SMG.	precuneus, PCG, Hypocampu
(Marcotte et al., 2018)	PCA	SFG, MTG, PrCG, Putamen, PrCG, SMA	PCG, PrCG, PoCG, precuneus Insula, ACG, IFG	SFG, STG, IFG,	Caudate, precuneus, STG, PrCG
(Nardo et al., 2017) (Fridriksson et al., 2006)	Phonemic cues Phonemic cues, naming			frontal pole, ACC, SMG, Temporal Pole, IPL	
(van Hees et al., 2014)	PCA	IFG (pars triangularis, opercularis), IFG (orbitalis), MTG, STG	IFG (pars triangularis, opercularis), MTG, STG	SMG, IFG (pars orbitalis and triangularis), AG,	AG
(Marcotte & Ansaldo, 2010)	SFA			SFG, IOG, globus pallidus	PrCG, SFG, IOG,
(Marcotte et al., 2012)	SFA			PrCG, IPL, MTG, IFG, SMG, SFG.	PrCG MTG, cuneus precuneus, STG, SFG,
(Kiran et al., 2015)	SFA			SFG, IFG, MFG, PoCG, SFG,	MFG, MTG, IFG,
(van Hees et al., 2014)	SFA	the IFG (pars orbitalis), FG (pars triangularis), and the MTG	MTG, STG, the IFG (pars opercularis)	AG, SMG, IFG (pars triangularis),	AG

ACC: anterior cingulate gyrus; AG: angular gyrus; FP: frontal pole; IOG: inferior occipital gyrus; IPL: Inferior parietal lobe; ITG: inferior temporal gyrus; MCG: middle cingulate gyrus; MOG: middle occipital gyrus; PaCL: paracentral lobule; PoCG: post central gyrus; PCG: posterior cingulate gyrus; SMA: supplementary motor area; SMG: supramarginal gyrus; STS: superior temporal sulcus; TP: temporal pole.

**Figure 2.** The rate of increased (red) and decreased (blue) activation of brain areas in the left hemisphere.

2010, 2012; Marcotte et al., 2012, 2018). In other words, standard protocols employ distributed bilateral areas and are associated with less naming improvement (Marcotte et al., 2018). In Menke et al's study (2009), intensive treatment of anomia therapy was associated with the significant improvement in the treated and moderate improvement of the untreated items. In this study, naming improvement was related to the recruitment of both hemisphere areas. Patients of this study have moderate-to-severe naming impairments, affecting neural plasticity mechanisms. Such effects were reported with changes limited to several language-related

areas in milder aphasia patients (van Hees et al., 2014) and the recruitment of bilateral brain areas in patients with moderate-to-severe word-finding difficulties (Marcotte et al., 2012; Menke et al., 2009). In addition to the severity of word-finding difficulties, impaired levels of word retrieval processing may also be related to response to different types of anomia therapies (Dignam et al., 2016; Kristinsson et al., 2021; Lambon Ralph et al., 2010; Lorenz & Ziegler, 2009). These variabilities may represent the various extent of activation in brain areas (Abel et al., 2014; Cornelissen et al., 2003; Fridriksson et al., 2012). For example, Marcotte and

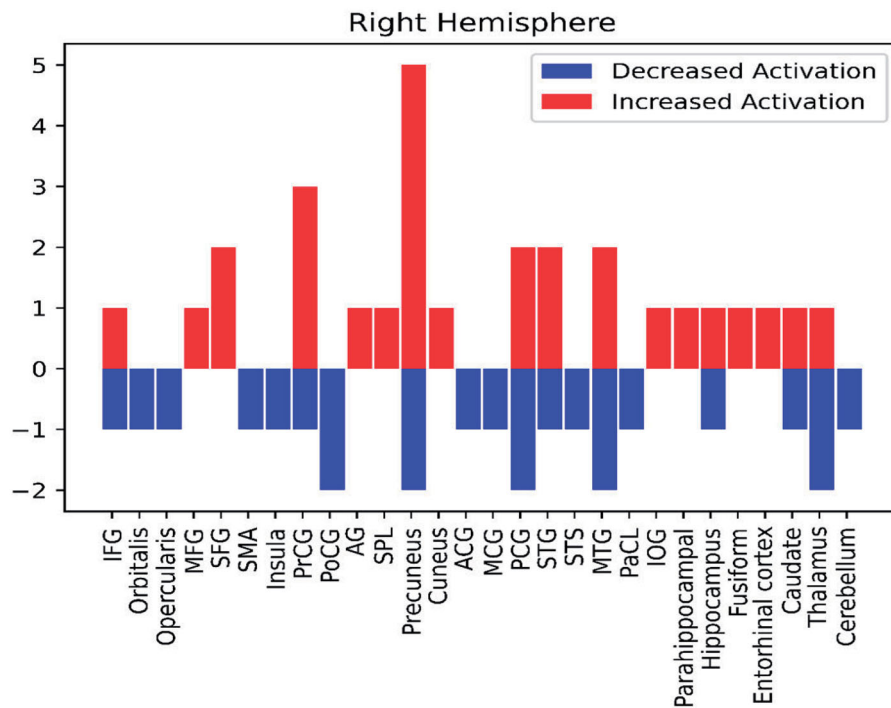


Figure 3. The rate of increased (red) and decreased (blue) activation of brain areas in the right hemisphere.

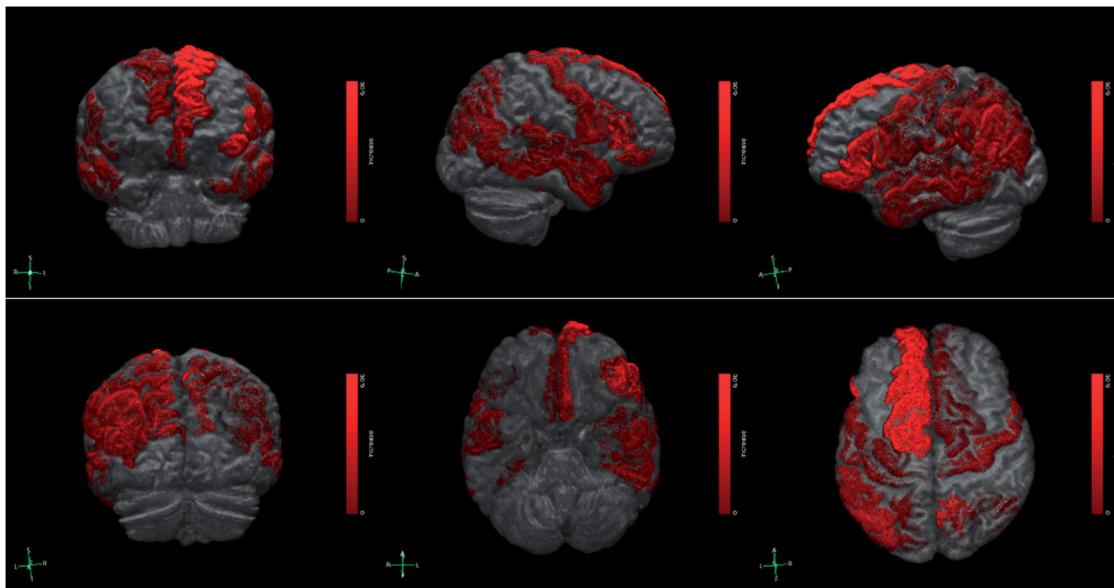


Figure 4. Mapping the effects of anomia therapy on brain structures; the ROIs that showed increased activation after anomia therapy are illustrated in the axial, sagittal, and coronal views.

Ansaldi (2010) showed that in patients with preserved semantic abilities, semantic treatment increased activation in areas related to speech production and articulatory planning. On the other hand, Abel et al. (2014) showed that in patients with impaired levels of phonological processing, phonemic cues increased activation in areas related to phonological processing (Abel et al., 2014). In sum, both hemispheres are associated with naming improvement; however, several variables may affect anomia therapy and, consequently, neural mechanisms of neuroplasticity.

Relation between decreased and increased activation and naming improvement

One crucial issue in investigating functional neuroplasticity is the relation between increased or decreased activation and treatment outcomes. Regarding anomia therapy, some studies reported that increased activation in language-related areas, the right homologs, perilesional areas, and areas that traditionally have not been related to language processing have a positive relation with anomia therapy (Fridriksson, 2010, 2007, 2012; Menke et al., 2009; Vitali et al., 2007). In

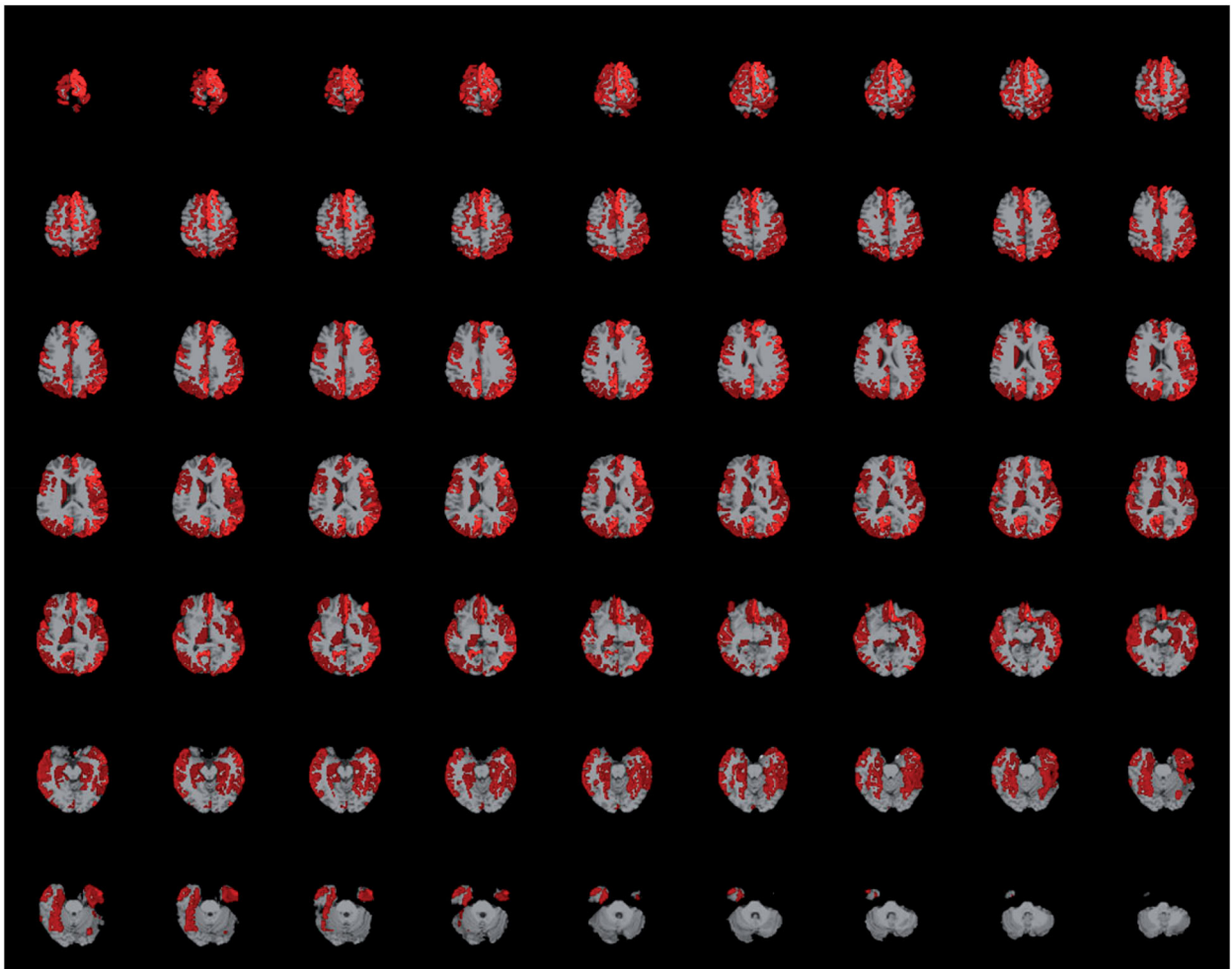


Figure 5. Sagittal slices (in the smaller size) demonstrate the distribution of the enhanced activation over the cortical areas.

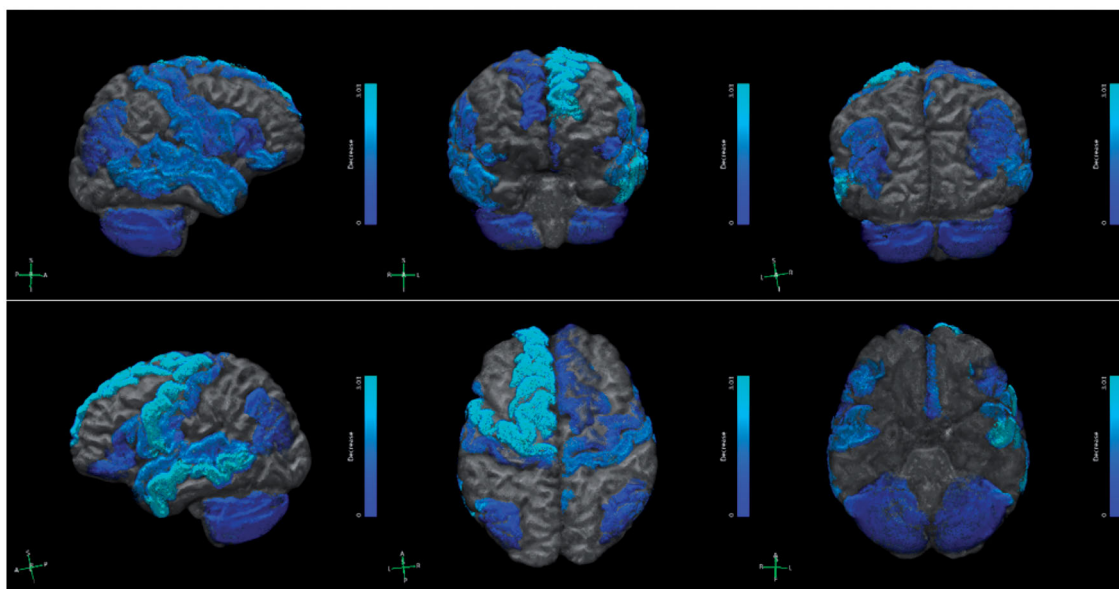


Figure 6. Mapping the effects of anomia therapy on brain structures; the ROIs that showed decreased activation after anomia therapy are illustrated in the axial, sagittal, and coronal views.

contrast, others showed that decreased activation in language-related areas and the right homologs is positively associated with treatment outcomes (Abel et al., 2014; 2015;

Marcotte et al., 2018; Nardo et al., 2017; van Hees et al., 2014). Focusing on semantic and phonological processing in improved word retrieval processing enhance the efficiency of

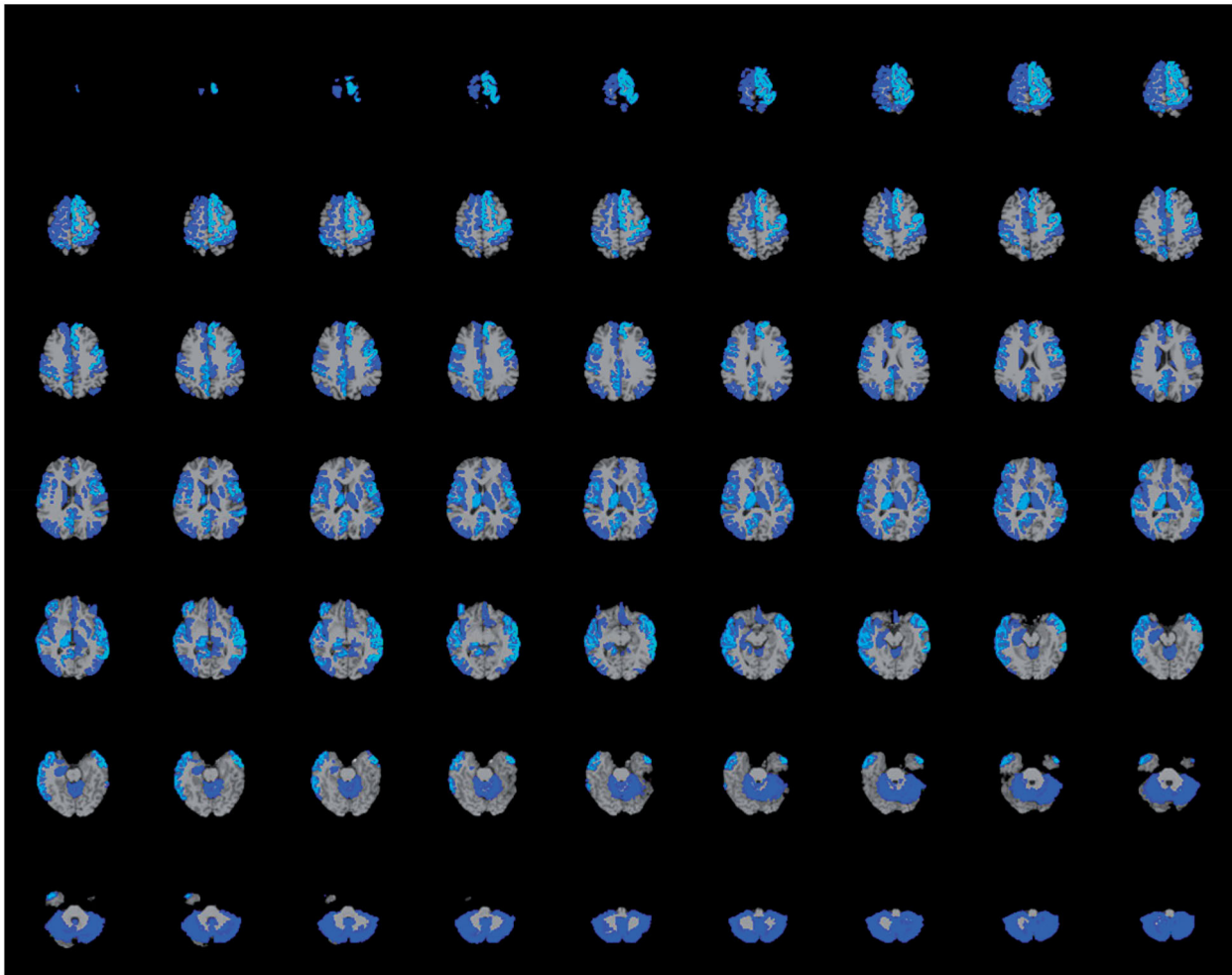


Figure 7. Sagittal slices (in the smaller size) demonstrate the distribution of the decreased activation of cortical areas.

language networks for a naming task that may explain the decreased activation of language-related areas after anomia therapy (Abel et al., 2015; Marcotte et al., 2018; Nardo et al., 2017; van Hees et al., 2014). The activation of other areas that traditionally are not related to language processing may have related to the effects of learning and memory consolidation processes on lexical items (Abel et al., 2014; Fridriksson et al., 2007; Marcotte et al., 2018; Menke et al., 2009). Extended lesion location may also affect brain activation. That is, larger and more paralyzing strokes cause the activation of those brain areas that are not initially involved in language (Abel et al., 2015; Fridriksson et al., 2007; Vitali et al., 2007).

Brain areas related to naming improvement

This review revealed that different brain areas might be related to anomia therapy-induced neuroplasticity. On the other hand, for naming a picture, an extensive and distributed network is involved throughout the frontal, parietal, temporal, and occipital lobes (Indefrey & Levelt, 2004; Nozari et al., 2020). Therefore, knowing about the language functioning of brain areas during word retrieval processing is a prerequisite for a better understanding of the mechanisms of anomia-therapy-induced neuroplasticity. This

section will discuss the potential relationship between anomia therapies and the rate of activation in brain areas.

Inferior frontal gyrus (IFG)

The activation of left IFG and its right homologs was associated with naming improvement after both semantic and phonemic-based anomia therapy (Abel et al., 2014; 2015; Fridriksson, 2010; 2007; Nardo et al., 2017; Vitali et al., 2007). Also, the damage to this area is negatively associated with anomia treatment (Abel et al., 2015; Marcotte et al., 2012). The critical role of this area in neuroplasticity induced by anomia therapy is probably due to the multiple subcomponents of this area in semantic and phonological processing of word production (Costafreda et al., 2006; Homae et al., 2002; Molnar-Szakacs et al., 2005; Newman et al., 2003; Nixon et al., 2004; Papathanassiou et al., 2000; Poldrack et al., 1999; Wright et al., 2011). The IFG contains Broca's area located in its posterior part (Nixon et al., 2004; Poldrack et al., 1999). Broca's area is composed of two sub-regions: the pars triangularis and the pars opercularis (Tomaiuolo et al., 1999). The opercular region (BA 44) of the IFG is necessary for the normal function of phonological working memory (Nixon et al., 2004) and is also involved in retrieving lexical or phonologic information during verbal

processing (Rosen et al., 2000). The relation between activation of the **pars opercularis after phonemic-based therapy is probably due to the role of this** subcomponent in phonological processing (Abel et al., 2014; Schwartz et al., 2012). Relation between anomia treatment and change in activation of pars triangularis during a semantic task has also been reported (Heath et al., 2012). Moreover, the negative association between the lesions of **pars triangularis with semantic treatment outcomes** (Marcotte et al., 2012) is probably because of the role of this area in semantic processing (Badre et al., 2005; Demb et al., 1995; Marcotte et al., 2012; Newman et al., 2003).

Middle frontal gyrus (MFG)

Studies reported increased activation of the left MFG during picture naming after phonological-based therapies (Abel et al., 2014; Fridriksson et al., 2012; van Hees et al., 2014; Vitali et al., 2007). Studies have shown the role of this area in the phonological storage of working memory (Heim et al., 2003), repetition tasks (Zhang et al., 2017), pre-articulation stage of word production (Wen et al., 2017), and verbal fluency (Dong et al., 2016). These data suggested that this area is involved in multiple cognitive functions, including the phonemic and executive stages of word retrieval processing. The activation of the MFG after aphasia treatment is mainly because of the role of this area in the phonetic aspects of retrieval processes (Cherney & Small, 2006; Fridriksson et al., 2012; Price, 2010; Vitali et al., 2007).

Superior frontal gyrus

This area is related to naming improvement induced by semantic and phonemic-based treatments (Fridriksson et al., 2007; Marcotte & Ansaldo, 2010; Nardo et al., 2017; van Hees et al., 2014). Other word-finding treatments also showed the activation of this area in phonological, semantic, and picture naming tasks (Kiran et al., 2015; Sandberg et al., 2015). This area included different subregions, including supplementary motor area (SMA) and pre-SMA (Li et al., 2013). The frontal aslant tract (FAT) is a short association fiber that connects the pre-SMA and SMA in SFG with IFG for motor execution (Dick et al., 2019; Szmuda et al., 2017; Wierenga et al., 2006). Therefore, it seems that the increased activation of SFG after aphasia therapy is due to the role of pre-SMA and SMA in the executive aspect of lexical production (Alario et al., 2006; Catani, 2019; Dick et al., 2019; Hertrich et al., 2016; Marcotte & Ansaldo, 2010; Nardo et al., 2017).

Precentral gyrus (PreCG)

The activation of the left (Marcotte et al., 2012; Marcotte & Ansaldo, 2010) and right preCG (Marcotte & Ansaldo, 2010) was reported after SFA therapy. The PreCG includes the primary motor and premotor cortices, and it has been associated with the programming and executive stages of word retrieval processing (Baldo et al., 2011; Itabashi et al.,

2016; Pulvermüller et al., 2006), explaining the activation of this area after anomia treatment (Marcotte & Ansaldo, 2010).

Cingulate cortex

Bilateral activation of the anterior (Fridriksson et al., 2006; Marcotte et al., 2012, 2018; Menke et al., 2009; Vitali et al., 2007) and posterior and midcingulate (Abel et al., 2014) gyri are associated with naming improvement. The cingulate cortex is located in the inner wall of each hemisphere (Paus, 2001) that is involved in executive functions and attentional processes needed for various cognitive tasks (MacDonald et al., 2000; Milham et al., 2001; Stamatakis et al., 2005; Vogt et al., 1992; Wager & Smith, 2003; Zhao et al., 2017). The role of the anterior cingulate in the initiation of speech and vocalization (Devinsky et al., 1995; Paus, 2001; Vogt et al., 1992), and also the involvement of posterior cingulate in the executive function aspect of semantic processing (Zhao et al., 2017) may represent the activation of this area in treatment-induced neuroplasticity (Abel et al., 2014; Fridriksson et al., 2006; Marcotte et al., 2018; Menke et al., 2009; Vitali et al., 2007).

Insula

The activation of this area was revealed after phonological-based therapies (Nardo et al., 2017; Vitali et al., 2007). The insula is located deep within the lateral sulcus of the brain, in the depths of the Sylvian fissure (Uddin et al., 2017). Insula broadens its connections to Broca's area, middle frontal gyrus, angular gyrus, Wernicke's area, supramarginal gyrus, SMA, and cingulate gyrus; therefore, it is involved in language processing. Due to the two latter connections (i.e., SMA and cingulate gyrus), damage to the insula may cause an inability to initiate and maintain voluntary speech production and mutism. Also, the disconnection of Broca's area and the insula in Broca's aphasia can cause apraxia and agrammatism (Ardila et al., 2014). Furthermore, the SMA and insula together constitute a word production motor network (Price et al., 1996) in which the left insula is engaged in speech planning and coordination (Mechelli et al., 2007). The activation of the insula after phonological training may imply the role of this area in articulatory planning, that is, the physical expression of word forms (Nardo et al., 2017; Vitali et al., 2007).

Inferior parietal lobe (IPL)

The activation of the inferior parietal lobe was seen in picture naming (Marcotte et al., 2012) after SFA therapy and semantic and phonological cueing treatments (Fridriksson, 2010; 2006; 2012). IPL is a heterogeneous region that includes essential areas for language processing: supramarginal (SMG) and angular gyrus (AG) (Stoeckel et al., 2009). Increased AG activation after SFA therapy was observed in picture naming (van Hees et al., 2014) and semantic decision (Kiran et al., 2015) tasks. Nardo et al. (2017) also

demonstrated that providing the lexical, phonological, and semantic forms of the target word increased the activation of this area. The AG takes a vital role in visualizing a concept as well as retrieval of words in all categories and is a prominent part of the human lexical/semantic network at the single-word level (Ardila et al., 2000; Binder et al., 2009; Bonner et al., 2013; Dronkers et al., 2004; Mechelli et al., 2007). Therefore, the language-related role of this area can be attributed to lexical/semantic processing (Crosson et al., 2005; Nardo et al., 2017; van Hees et al., 2014).

SMG activation was associated with naming improvement after phonemic-based therapy (van Hees et al., 2014; Vitali et al., 2007). Also, Rochon et al. (2010) showed the increased activation of SMG in a phonological judgment task. The primary language-related function of this area is related to phonological word-formation processing (Deschamps et al., 2014; Oberhuber et al., 2016; Sliwiska et al., 2012; Stoeckel et al., 2009). As a result, it appears that the SMG is the center for phonological processing, giving it a vital role in phonological processing aspects of word production (Cornelissen et al., 2003; van Hees et al., 2014; Vitali et al., 2007).

Superior temporal gyrus (STG)

Activation of the STG after anomia therapy is reported by (Abel et al., 2014; Marcotte & Ansaldi, 2010; van Hees et al., 2014). Increased activation of this area was also reported in auditory-speech processing tasks (Mohr et al., 2014) and oral comprehension tasks after language therapy (Mattoli et al., 2014). The STG is the central area for auditory processing and analyzes acoustic signals such as speech (Howard et al., 2000; Visser & Lambon Ralph, 2011). Therefore, it is involved in auditory comprehension processing (Chang et al., 2010; Dronkers et al., 2004; Leff et al., 2009; Mesgarani et al., 2014). Also, according to the motor control theory of speech production, the superior temporal gyrus bilaterally evaluates the auditory signals of articulatory movements. When an articulatory error during spoken language is noticed, the right STG sends signals to the right premotor cortex to correct the error (Tourville et al., 2008). Although a few studies reported increased activation of the STG after anomia therapy, this area's role in word retrieval is probably related to auditory and feedback processing of word production (Abel et al., 2014).

Middle temporal gyrus

Studies that utilized both semantic and phonemic cues showed that changes in the activation of this area are associated with naming improvement (Abel et al., 2014; 2015; Fridriksson et al., 2007; Marcotte & Ansaldi, 2010; van Hees et al., 2014). The increased activation of this area was also seen in different types of aphasia therapy with different language tasks (Abel et al., 2014; Gili et al., 2017; Kiran et al., 2015; Kurland et al., 2012; Meinzer et al., 2007; Mohr et al., 2014; Richter et al., 2008; Thompson et al., 2010; 2013; Wierenga et al., 2006; Zhang et al., 2017; Zipse et al.,

2012), that shows this area's involvement in nearly all language tasks (Dronkers et al., 2004). This region is a "common semantic system" for word and object stimuli (Binder & Desai, 2011; Maess et al., 2002) that is involved in the human lexical and semantic network of lexical processing (Binder et al., 2009; Heath et al., 2012; Indefrey & Levelt, 2000; Lukic et al., 2017; Price, 2010; Python et al., 2018; Smith et al., 2013; Toichadse et al., 2018; Whitney et al., 2011). Activation of this area during semantic and phonological errors also emphasizes the role of this area in lexical retrieval processing (Abel et al., 2015; Fridriksson et al., 2012).

Inferior temporal gyrus (ITG)

Some studies revealed the activation of both the right and left ITG in picture naming tasks after SFA therapy (Fridriksson et al., 2007; Marcotte & Ansaldi, 2010; van Hees et al., 2014).

The inferior temporal gyrus is a part of the ventral occipitotemporal (VOT) region, which is involved in object and face recognition (Borghesani et al., 2019; Grill-Spector, 2003). The posterior ITG involvement in lexical decision tasks (Dien et al., 2013) and the early stages of writing (Potgieser et al., 2015) also shows the role of this area in word recognition. One study demonstrated the activation of this area in semantic tasks after SFA therapy (Kiran et al., 2015), which agrees with this area's role in word recognition. It seems that increased activation of the ITG after aphasia therapy is due to its role in picture recognition.

Parahippocampal gyrus (PG)

This area's activation was also observed after phonological (Menke et al., 2009; Vitali et al., 2007) and semantic-based treatments (van Hees et al., 2014). The PG is the main structure of semantic cognition (Ding et al., 2016; Ricci et al., 1999; Toichadse et al., 2018) and is an essential structure for learning and memory (Bartha et al., 2003; Price et al., 1996; Squire et al., 2004). The relationship between this area's activation and anomia therapy probably consolidates word forms in the memory (Fridriksson et al., 2007; Menke et al., 2009; van Hees et al., 2014; Vitali et al., 2007).

Precuneus

Studies results showed the involvement of the precuneus in picture naming tasks after semantic and phonemic-based therapies (Abel et al., 2014; Fridriksson, 2007, 2010; Menke et al., 2009; van Hees et al., 2014). Precuneus is located in the posteromedial cortex of the parietal lobe (Brodmann area 7) (Cavanna & Trimble, 2006) and is involved in phonemic (Voets et al., 2006) and semantic fluency tasks (Nenert et al., 2018). Both phonemic and semantic fluencies represent the ability of retrieval capacities and the initiation of searches in subcategories (Cavaco et al., 2013; Gourovitch et al., 2000; Kavé, 2005; Vignando et al., 2019). Because precuneus is frequently activated after various therapies

(Crosson et al., 2005; Harnish et al., 2008; Heath et al., 2015; Kurland et al., 2012), it seems that this area is activated in the basic levels of language processing; probably in the recognition stage of word retrieval (Vignando et al., 2019). Furthermore, it is believed that the effects of semantic and phonemic priming are mediated by episodic memory (Mitchell & Brown, 1988; Tenpenny, 1995), and the precuneus is involved in episodic memory retrieval (Cavanna & Trimble, 2006). The activation of this area after phonemic priming during the naming (Heath et al., 2015) is also related to the area's role in memory retrieval (Heath et al., 2015; Rochon et al., 2010; van Hees et al., 2014).

Subcortical structures

Thalamus

According to the cortico-striato-thalamocortical language processing, the thalamus monitors the interactions between fronto-opercular and temporo-cortical cortices to integrate lexico-syntactic and semantic information. At the same time, basal ganglia inhibit the error of a similar but unrelated word formation (e.g., CAT instead of HAT) (Klostermann et al., 2013). Increased left cerebellar and right thalamic activity following the phonological treatment can reflect the increased reliance on motor feedback to predict naming errors (Abel et al., 2014; Fridriksson et al., 2007; Nasir & Ostry, 2006).

Caudate

Several studies reported the association between the activation of the caudate and naming therapy outcome (Abel et al., 2014; Marcotte et al., 2018; van Hees et al., 2014). This area is involved in language control by suppressing irrelevant words (Abutalebi & Green*, 2008; Duffau et al., 2014; Robles et al., 2005), associative memory (Wallentin et al., 2006), episodic memory retrieval (Assaf et al., 2006; Hart et al., 2013), and procedural learning of language tasks are other functions of this area (Ullman, 2004). Therefore, selecting the most relevant language features and learning new words occur following the activation of this area after anomia therapy (Abel et al., 2014; Marcotte et al., 2018; van Hees et al., 2014).

Cerebellum

The cerebellum has structural and functional connectivities with the cortex. The disconnection of the cerebellum hemispheres causes various symptoms such as amnesia and different types of aphasia, including agrammatism and transcortical-motor aphasia. This network also appears to be engaged in motor control and executive functions (e.g., subvocal rehearsal mechanisms of verbal working memory) and seems to be used during distinct speech perception tasks (Ackermann et al., 2007). In addition to motor control, the cerebellum has a role in the timing, sequencing, processing, and sensorimotor adaptation of high-level cognitive modalities. In verbal working memory, the superior cerebellar lobule VI and crus I are involved in the initiation of

transforming phonological content to motor performance during information encoding. The cerebellum has a role in verbal fluency (both semantic and phonemic), and this supports the hypothesis that sequence processing is the primary function of the cerebellum in the language (Mariën et al., 2014). The increased activation of the cerebellum in picture naming after PCA therapy (Fridriksson et al., 2007; van Hees et al., 2014) and melodic intonation therapy (Zipse et al., 2012), also is in line with its role in phonemic processing and executive function of language system. It has been shown that the left cerebellum is activated while naming an object and hence is probably involved in object recognition (Price et al., 1996). Several treatments (Fridriksson et al., 2007; Marcotte et al., 2013; van Hees et al., 2014; Zipse et al., 2012) showed the cerebellum activation in the picture naming task, which also may be due to the role of this area in object recognition.

Limitations

The semantic and phonemic cueing treatments are two common anomia treatments that affect all word retrieval processing levels. However, because semantic-based treatments concentrate on semantic processing and phonemic-based treatments focus on phonological processing, these two treatments probably employ different mechanisms. Most studies alternately used these two treatments, and therefore it is hardly challenging that dissociate the effects of these two treatments on brain function. Also, the design of the studies was different. Some studies used a single-subject design, and others reported group analyses. These factors can cause diversity in the results. The systematic integration of different study results might control this variability. However, the lack of reports about the peak locations of treatment-related effects in these heterogeneous studies makes meta-analysis impossible. This is a significant limitation, and future research must focus on it because meta-analyses could integrate variable results from the small-data studies, which are still very frequent in this field of research.

Conclusion

Studies have shown several mechanisms for neuroplasticity induced by anomia therapies. Sometimes increased activation in language, motor, learning, and memory-related areas are associated with naming improvements. In the other cases, decreased activation in the left and right homologs of language-related areas may be related to naming improvement. The extended lesion in core language-related areas, the severity of word-finding difficulties, the modality chosen as the treatment's goal, and the levels of impairment in word retrieval processing are associated with mechanisms of neuroplasticity. Because the studies showed that left hemisphere reactivation is associated with more naming improvements, intensive treatment that increases left hemisphere recruitment for anomia therapy could be an effective protocol in anomia practice.

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